Chapter 8. Functions Defined by Integrals

Continuity and differentiation under the integral sign, the gamma and beta functions, integral transforms

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Many important functions are defined as integrals depending on a parameter, $\phi(x) = \int f(x,t)\,dt$. We study when such ϕ is continuous or differentiable, and then meet the gamma function and integral transforms.

8.1 Continuity and differentiation under the integral sign

Proposition 8.1

If f is continuous on $[a,b]\times[c,d]$, then $\phi(x)=\int_c^d f(x,t)dt$ is continuous on $[a,b]$. If moreover $\partial_x f$ exists and is continuous on $[a,b]\times[c,d]$, then ϕ is C^1 and $\phi'(x)=\int_c^d \partial_x f(x,t)dt$.

Proof. f is uniformly continuous on the compact rectangle, so $|f(x,t)-f(x_0,t)|<\epsilon$ for all t when $|x-x_0|<\delta$; integrate. For the derivative, by the mean value theorem $\frac{f(x+h,t)-f(x,t)}{h}=\partial_x f(x+\theta h,t)$, which converges to $\partial_x f(x,t)$ uniformly in t by uniform continuity of $\partial_x f$; apply Proposition 6.5. ∎

Integrating ϕ over $[a,b]$ and using this result with $\Phi(y)=\int_c^d\int_a^y f(x,t)dx\,dt$ proves **Fubini's theorem for continuous functions**: $\int_a^b\int_c^d f\,dt\,dx=\int_c^d\int_a^b f\,dx\,dt$. (Differentiate both sides in the upper limit y.)

For improper integrals in t we need uniformity. $\int_c^\infty f(x,t)dt$ **converges uniformly** on A if $\sup_{x\in A}|\int_T^\infty f(x,t)dt|\to 0$ as $T\to\infty$.

Theorem 8.2 (Dominated improper integrals)

Let f be continuous on $[a,b]\times[c,\infty)$ with $|f(x,t)|\le g(t)$ and $\int_c^\infty g<\infty$. Then $\phi(x)=\int_c^\infty f(x,t)dt$ converges uniformly and is continuous on $[a,b]$.

Proof. $|\phi(x)-\int_c^T f(x,t)dt|\le\int_T^\infty g\to 0$ uniformly in x, and each truncated integral is continuous by Proposition 8.1; apply Theorem 6.1. ∎

Theorem 8.3 (Differentiation of improper integrals)

Let f and $\partial_x f$ be continuous on $(a,b)\times[c,\infty)$, $\int_c^\infty f(x_0,t)dt$ converge for some x_0, and $\int_c^\infty \partial_x f(x,t)dt$ converge uniformly on (a,b). Then $\phi(x)=\int_c^\infty f(x,t)dt$ converges, is differentiable, and $\phi'(x)=\int_c^\infty \partial_x f(x,t)dt$.

Proof. Apply Theorem 6.4 to $\phi_T(x)=\int_c^T f(x,t)dt$ as $T\to\infty$ (along any sequence), using Proposition 8.1 for ϕ_T'. ∎

Example 8.1

(a) The Gaussian integral: $I=\int_0^\infty e^{-t^2}dt=\frac{\sqrt{\pi}}{2}$. Let $F(x)=\left(\int_0^x e^{-t^2}dt\right)^2$ and $G(x)=\int_0^1\frac{e^{-x^2(1+t^2)}}{1+t^2}dt$. Differentiating (Proposition 8.1) and substituting $s=xt$ gives $F'+G'=0$, so $F+G=G(0)=\frac{\pi}{4}$; letting $x\to\infty$, $I^2=\frac{\pi}{4}$.
(b) The Dirichlet integral: $\phi(x)=\int_0^\infty e^{-xt}\frac{\sin t}{t}dt$ for $x>0$ has $\phi'(x)=-\int_0^\infty e^{-xt}\sin t\,dt=-\frac{1}{1+x^2}$ (the derivative integral converges uniformly on $x\ge\delta>0$), and $\phi(x)\to 0$ as $x\to\infty$, so $\phi(x)=\frac{\pi}{2}-\arctan x$. Uniform convergence on $[0,\infty)$ (via integration by parts) lets $x\to 0+$: $\int_0^\infty\frac{\sin t}{t}dt=\frac{\pi}{2}$.

8.2 The gamma function

Definition (Gamma and beta functions)

$\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}dt$ for $x > 0$, and $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$ for $x, y > 0$.

Both integrals converge (compare with t^{x-1} near 0 and $e^{-t/2}$ at infinity). By Theorems 8.2–8.3 on compact subintervals of $(0,\infty)$, Γ is C^∞ with $\Gamma^{(k)}(x) = \int_0^\infty (\log t)^k t^{x-1}e^{-t}dt$. Integration by parts gives $\Gamma(x+1) = x\Gamma(x)$, and $\Gamma(1) = 1$, so $\Gamma(n+1) = n!$. Also $\Gamma(\tfrac12) = 2\int_0^\infty e^{-s^2}ds = \sqrt{\pi}$.

A positive function is **log-convex** if $\log f$ is convex. By Hölder's inequality ($\int uv \le (\int u^p)^{1/p}(\int v^q)^{1/q}$, $\frac1p + \frac1q = 1$) applied to $t^{x/p-1/p}e^{-t/p}$ and $t^{y/q-1/q}e^{-t/q}$,

$$\Gamma\Big(\frac{x}{p} + \frac{y}{q}\Big) \le \Gamma(x)^{1/p}\Gamma(y)^{1/q},$$

so Γ is log-convex. A continuous function satisfying the midpoint inequality $f(\frac{x+y}{2}) \le \frac{f(x)+f(y)}{2}$ is convex (the inequality extends to dyadic weights by induction, and to all weights by continuity), which is often an easier way to verify convexity.

Theorem 8.4 (Bohr–Mollerup)

Γ is the only function $f : (0,\infty) \to (0,\infty)$ such that (a) $f(x+1) = xf(x)$, (b) $f(1) = 1$, (c) $\log f$ is convex. Moreover, for $x > 0$,

$$\Gamma(x) = \lim_{n\to\infty} \frac{n!\,n^x}{x(x+1)\cdots(x+n)}.$$

Proof. Let f satisfy (a)–(c) and $\varphi = \log f$. By (a), (b) it suffices to treat $0 < x < 1$. For integers n, convexity bounds the slope of φ on $[n, n+x]$ between the slopes on $[n-1, n]$ and $[n, n+1]$, which are $\log(n-1)$ and $\log n$:

$$x\log(n-1) \le \varphi(n+x) - \varphi(n) \le x\log n.$$

Since $\varphi(n) = \log(n-1)!$ and $\varphi(n+x) = \varphi(x) + \log\{x(x+1)\cdots(x+n-1)\}$, this gives $\frac{(n-1)!(n-1)^x}{x(x+1)\cdots(x+n-1)} \le f(x) \le \frac{(n-1)!\,n^x}{x(x+1)\cdots(x+n-1)}$. Both bounds tend to the stated limit (their ratio tends to 1), so f is determined, and Γ satisfies (a)–(c). ∎

Corollary 8.5

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

Proof. Fix y and let $f(x) = \Gamma(x+y)B(x,y)/\Gamma(y)$. Integration by parts gives $B(x+1,y) = \frac{x}{x+y}B(x,y)$, so $f(x+1) = xf(x)$; $B(1,y) = 1/y$ gives $f(1) = 1$; and $B(\cdot,y)$ is log-convex by Hölder, as is $\Gamma(\cdot + y)$. By Bohr–Mollerup, $f = \Gamma$. ∎

Example 8.2

(a) $\int_0^{\pi/2} \sin^{2x-1}\theta\cos^{2y-1}\theta\,d\theta = \frac12 B(x,y)$ (substitute $t = \sin^2\theta$); e.g. $\int_0^{\pi/2}\sin^n\theta\,d\theta = \frac{\sqrt{\pi}}{2}\frac{\Gamma((n+1)/2)}{\Gamma(n/2+1)}$.

(b) The volume of the unit ball in $\mathbb{R}^n$ is $\pi^{n/2}/\Gamma(\frac{n}{2}+1)$ (compute $\int_{\mathbb{R}^n} e^{-\|x\|^2}dx = \pi^{n/2}$ in two ways).

(c) Stirling's formula $\Gamma(x+1) \sim \sqrt{2\pi x}\, x^x e^{-x}$ follows by writing $\Gamma(x+1) = x^{x+1}e^{-x}\int_0^\infty e^{-x(s-1-\log s)}ds$ and noting that $s-1-\log s \approx (s-1)^2/2$ near $s=1$ (Laplace's method).

8.3 Integral transforms

A function of the form $F(x) = \int_I k(x,t)f(t)dt$ is an **integral transform** of f with **kernel** k. Gamma and beta are of this type with $k(x,t) = t^{x-1}$ (the Mellin transform), and the Hölder argument above shows that $\log F$ is convex whenever $f \geq 0$ and F is continuous.

The **Laplace transform** is $(\mathcal{L}f)(x) = \int_0^\infty e^{-xt}f(t)dt$, defined for x large when, e.g., $|f(t)| \leq Ce^{at}$. Direct computation gives

$$\mathcal{L}[e^{at}](x) = \frac{1}{x-a}, \quad \mathcal{L}[\cos at](x) = \frac{x}{x^2+a^2}, \quad \mathcal{L}[\sin at](x) = \frac{a}{x^2+a^2} \qquad (x > a,\ x > 0),$$

the last two from the first with a replaced by $\pm ia$. If f is C^1 and f, f' grow at most exponentially, integration by parts gives

$$\mathcal{L}[f'](x) = x\,\mathcal{L}[f](x) - f(0),$$

which turns linear differential equations with constant coefficients into algebraic equations.

The **Fourier transform** is $\hat{f}(x) = \int_{-\infty}^\infty e^{-itx}f(t)dt$, defined for every x when $\int|f| < \infty$. The Laplace transform is the special case of complex argument: with $x = u+iv$, $\int_0^\infty e^{-tx}f = \int_0^\infty e^{-itv}e^{-tu}f(t)dt$. By Theorem 8.2, $\hat{f}$ is continuous and $|\hat{f}| \leq \int|f|$; the Riemann–Lebesgue lemma of Chapter 9 shows $\hat{f}(x) \to 0$ as $|x| \to \infty$. For example, $f = \mathbf{1}_{[-1,1]}$ gives $\hat{f}(x) = \frac{2\sin x}{x}$, and $f(t) = e^{-t^2/2}$ gives $\hat{f}(x) = \sqrt{2\pi}e^{-x^2/2}$ (differentiate under the integral to get $\hat{f}' = -x\hat{f}$).